```
# ========================================
# Spark for Machine Learning - Price Prediction
# Linear Regression on Airbnb Dataset
# ========================================

# Step 0: Install and Start Spark

!pip install pyspark
```

```
Requirement already satisfied: pyspark in /usr/local/lib/python3.12/dist-packages (4.0.2)
Requirement already satisfied: py4j<0.10.9.10,>=0.10.9.7 in /usr/local/lib/python3.12/dist-packages (from pyspark) (0.10.9.9)
```

```
from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .appName("Airbnb Price Prediction") \
    .getOrCreate()

print("Spark started successfully")
```

```
Spark started successfully
```

```
from google.colab import files

uploaded = files.upload()
```

Choose Files airbnb_pric…_sample.csv

**airbnb_price_prediction_sample.csv**(text/csv) - 8360 bytes, last modified: 5/9/2026 - 100% done

```
Saving airbnb_price_prediction_sample.csv to airbnb_price_prediction_sample.csv
```

```
df = spark.read.csv(
    "airbnb_price_prediction_sample.csv",
    header=True,
    inferSchema=True
)

df.show(5)
```

```
+---+---------+--------+----+------------+--------------+-----------------+--------------------+-------------+------------+--
| id|bathrooms|bedrooms|beds|accommodates|minimum_nights|number_of_reviews|review_scores_rating|property_type|   room_type|
+---+---------+--------+----+------------+--------------+-----------------+--------------------+-------------+------------+--
|  1|      1.0|     1.0| 2.0|           3|             2|               57|                74.2|    Apartment| Shared room|Ga
|  2|      3.0|     1.0| 1.0|           2|             2|               59|                85.2|    Apartment| Shared room|
|  3|      3.0|     1.0| 1.0|           4|             3|              207|                96.1|        House| Shared room|
|  4|      1.0|     1.0| 1.0|           2|             1|               97|                72.9| Private room| Shared room|
|  5|      4.0|     3.0| 3.0|           7|             1|              141|                78.8|       Studio|Private room|Ga
+---+---------+--------+----+------------+--------------+-----------------+--------------------+-------------+------------+--
only showing top 5 rows
```

```
df.printSchema()
```

```
root
 |-- id: integer (nullable = true)
 |-- bathrooms: double (nullable = true)
 |-- bedrooms: double (nullable = true)
 |-- beds: double (nullable = true)
 |-- accommodates: integer (nullable = true)
 |-- minimum_nights: integer (nullable = true)
 |-- number_of_reviews: integer (nullable = true)
 |-- review_scores_rating: double (nullable = true)
 |-- property_type: string (nullable = true)
 |-- room_type: string (nullable = true)
 |-- neighborhood: string (nullable = true)
 |-- price: double (nullable = true)
```

```
df.select("bathrooms", "bedrooms", "price").show()
```

```
+---------+--------+------+
|bathrooms|bedrooms| price|
+---------+--------+------+
```

```
|      1.0|     1.0|146.65|
|      3.0|     1.0| 252.6|
|      3.0|     1.0|274.16|
|      1.0|     1.0|168.71|
|      4.0|     3.0|431.81|
|      1.0|     1.0|195.22|
|      2.5|     1.0|227.21|
|      1.5|     1.0|259.44|
|      1.0|     1.0|149.63|
|      4.0|     2.0|363.65|
|      3.0|     3.0|429.59|
|      1.0|     4.0|450.65|
|      1.0|     4.0|429.74|
|      4.0|     1.0|319.03|
|      2.0|     4.0|482.72|
|      2.5|     2.0|346.94|
|      1.0|     1.0|252.54|
|      1.0|     2.0|272.25|
|      3.0|     4.0|504.45|
|      1.0|     3.0|362.57|
+---------+--------+------+
only showing top 20 rows
```

```
df.select("bathrooms", "bedrooms", "price").summary().show()
```

```
+-------+------------------+------------------+-----------------+
|summary|         bathrooms|          bedrooms|            price|
+-------+------------------+------------------+-----------------+
|  count|               120|               120|              120|
|   mean|2.0791666666666666| 2.183333333333333|331.6846666666669|
| stddev|1.1320314032919827|1.0999108698292912|99.93166069855937|
|    min|               1.0|               1.0|           144.23|
|    25%|               1.0|               1.0|           259.96|
|    50%|               1.5|               2.0|           318.44|
|    75%|               3.0|               3.0|           408.92|
|    max|               4.0|               4.0|           565.58|
+-------+------------------+------------------+-----------------+
```

```
df_clean = df.na.drop(subset=["bathrooms", "bedrooms", "price"])

df_clean.select("bathrooms", "bedrooms", "price").show()
```

```
+---------+--------+------+
|bathrooms|bedrooms| price|
+---------+--------+------+
|      1.0|     1.0|146.65|
|      3.0|     1.0| 252.6|
|      3.0|     1.0|274.16|
|      1.0|     1.0|168.71|
|      4.0|     3.0|431.81|
|      1.0|     1.0|195.22|
|      2.5|     1.0|227.21|
|      1.5|     1.0|259.44|
|      1.0|     1.0|149.63|
|      4.0|     2.0|363.65|
|      3.0|     3.0|429.59|
|      1.0|     4.0|450.65|
|      1.0|     4.0|429.74|
|      4.0|     1.0|319.03|
|      2.0|     4.0|482.72|
|      2.5|     2.0|346.94|
|      1.0|     1.0|252.54|
|      1.0|     2.0|272.25|
|      3.0|     4.0|504.45|
|      1.0|     3.0|362.57|
+---------+--------+------+
only showing top 20 rows
```

```
trainDF, testDF = df_clean.randomSplit([0.8, 0.2], seed=42)

print("Training Rows:", trainDF.count())
print("Testing Rows:", testDF.count())
```

```
Training Rows: 98
Testing Rows: 22
```

```
from pyspark.ml.feature import VectorAssembler

vecAssembler = VectorAssembler(
    inputCols=["bathrooms", "bedrooms"],
    outputCol="features"
)
```

```
vecTrainDF = vecAssembler.transform(trainDF)

vecTrainDF.select(
    "bathrooms",
    "bedrooms",
    "features",
    "price"
).show()
```

```
+---------+--------+---------+------+
|bathrooms|bedrooms| features| price|
+---------+--------+---------+------+
|      1.0|     1.0|[1.0,1.0]|146.65|
|      3.0|     1.0|[3.0,1.0]| 252.6|
|      1.0|     1.0|[1.0,1.0]|168.71|
|      4.0|     3.0|[4.0,3.0]|431.81|
|      1.0|     1.0|[1.0,1.0]|195.22|
|      1.5|     1.0|[1.5,1.0]|259.44|
|      4.0|     2.0|[4.0,2.0]|363.65|
|      3.0|     3.0|[3.0,3.0]|429.59|
|      1.0|     4.0|[1.0,4.0]|450.65|
|      1.0|     4.0|[1.0,4.0]|429.74|
|      2.0|     4.0|[2.0,4.0]|482.72|
|      2.5|     2.0|[2.5,2.0]|346.94|
|      1.0|     1.0|[1.0,1.0]|252.54|
|      1.0|     2.0|[1.0,2.0]|272.25|
|      3.0|     4.0|[3.0,4.0]|504.45|
|      2.5|     1.0|[2.5,1.0]|262.21|
|      1.5|     2.0|[1.5,2.0]|283.55|
|      1.0|     4.0|[1.0,4.0]|447.73|
|      1.0|     1.0|[1.0,1.0]|199.55|
|      3.0|     2.0|[3.0,2.0]|335.79|
+---------+--------+---------+------+
only showing top 20 rows
```

```
from pyspark.ml.regression import LinearRegression

lr = LinearRegression(
    featuresCol="features",
    labelCol="price",
    predictionCol="prediction"
)
```

```
lrModel = lr.fit(vecTrainDF)

print("Model trained successfully")
```

```
Model trained successfully
```

```
predDF = lrModel.transform(vecTrainDF)

predDF.select(
    "bathrooms",
    "bedrooms",
    "features",
    "price",
    "prediction"
).show()
```

```
+---------+--------+---------+------+------------------+
|bathrooms|bedrooms| features| price|        prediction|
+---------+--------+---------+------+------------------+
|      1.0|     1.0|[1.0,1.0]|146.65|205.81981563880692|
|      3.0|     1.0|[3.0,1.0]| 252.6| 269.0154678055155|
|      1.0|     1.0|[1.0,1.0]|168.71|205.81981563880692|
|      4.0|     3.0|[4.0,3.0]|431.81| 459.4643705517651|
|      1.0|     1.0|[1.0,1.0]|195.22|205.81981563880692|
|      1.5|     1.0|[1.5,1.0]|259.44|221.61872868048408|
|      4.0|     2.0|[4.0,2.0]|363.65|380.03883222031743|
```

```
|      3.0|     3.0|[3.0,3.0]|429.59| 427.8665444684108|
|      1.0|     4.0|[1.0,4.0]|450.65| 444.0964306331499|
|      1.0|     4.0|[1.0,4.0]|429.74| 444.0964306331499|
|      2.0|     4.0|[2.0,4.0]|482.72| 475.6942567165041|
|      2.5|     2.0|[2.5,2.0]|346.94|  332.642093095286|
|      1.0|     1.0|[1.0,1.0]|252.54|205.81981563880692|
|      1.0|     2.0|[1.0,2.0]|272.25|285.24535397025454|
|      3.0|     4.0|[3.0,4.0]|504.45|507.29208279985846|
|      2.5|     1.0|[2.5,1.0]|262.21|253.21655476383833|
|      1.5|     2.0|[1.5,2.0]|283.55|301.04426701193177|
|      1.0|     4.0|[1.0,4.0]|447.73| 444.0964306331499|
|      1.0|     1.0|[1.0,1.0]|199.55|205.81981563880692|
|      3.0|     2.0|[3.0,2.0]|335.79|348.44100613696315|
+---------+--------+---------+------+------------------+
only showing top 20 rows
```

```
vecTestDF = vecAssembler.transform(testDF)

vecTestDF.select(
    "bathrooms",
    "bedrooms",
    "features",
    "price"
).show()
```

```
+---------+--------+---------+------+
|bathrooms|bedrooms| features| price|
+---------+--------+---------+------+
|      3.0|     1.0|[3.0,1.0]|274.16|
|      2.5|     1.0|[2.5,1.0]|227.21|
|      1.0|     1.0|[1.0,1.0]|149.63|
|      4.0|     1.0|[4.0,1.0]|319.03|
|      1.0|     3.0|[1.0,3.0]|362.57|
|      4.0|     1.0|[4.0,1.0]|255.15|
|      2.0|     1.0|[2.0,1.0]|217.86|
|      4.0|     3.0|[4.0,3.0]|445.83|
|      3.0|     2.0|[3.0,2.0]|336.57|
|      1.0|     3.0|[1.0,3.0]|362.99|
|      2.5|     2.0|[2.5,2.0]|302.99|
|      4.0|     3.0|[4.0,3.0]|452.07|
|      4.0|     1.0|[4.0,1.0]|284.39|
|      3.0|     4.0|[3.0,4.0]|494.39|
|      1.0|     1.0|[1.0,1.0]| 169.1|
|      4.0|     3.0|[4.0,3.0]|457.96|
|      4.0|     2.0|[4.0,2.0]|377.29|
|      1.0|     2.0|[1.0,2.0]|272.12|
|      1.0|     2.0|[1.0,2.0]|274.42|
|      1.0|     2.0|[1.0,2.0]|267.16|
+---------+--------+---------+------+
only showing top 20 rows
```

```
predTestDF = lrModel.transform(vecTestDF)

predTestDF.select(
    "bathrooms",
    "bedrooms",
    "features",
    "price",
    "prediction"
).show()
```

```
+---------+--------+---------+------+------------------+
|bathrooms|bedrooms| features| price|        prediction|
+---------+--------+---------+------+------------------+
|      3.0|     1.0|[3.0,1.0]|274.16| 269.0154678055155|
|      2.5|     1.0|[2.5,1.0]|227.21|253.21655476383833|
|      1.0|     1.0|[1.0,1.0]|149.63|205.81981563880692|
|      4.0|     1.0|[4.0,1.0]|319.03|300.61329388886975|
|      1.0|     3.0|[1.0,3.0]|362.57| 364.6708923017022|
|      4.0|     1.0|[4.0,1.0]|255.15|300.61329388886975|
|      2.0|     1.0|[2.0,1.0]|217.86|237.41764172216122|
|      4.0|     3.0|[4.0,3.0]|445.83| 459.4643705517651|
|      3.0|     2.0|[3.0,2.0]|336.57|348.44100613696315|
|      1.0|     3.0|[1.0,3.0]|362.99| 364.6708923017022|
|      2.5|     2.0|[2.5,2.0]|302.99|  332.642093095286|
|      4.0|     3.0|[4.0,3.0]|452.07| 459.4643705517651|
|      4.0|     1.0|[4.0,1.0]|284.39|300.61329388886975|
|      3.0|     4.0|[3.0,4.0]|494.39|507.29208279985846|
|      1.0|     1.0|[1.0,1.0]| 169.1|205.81981563880692|
|      4.0|     3.0|[4.0,3.0]|457.96| 459.4643705517651|
```

```
|      4.0|     2.0|[4.0,2.0]|377.29|380.03883222031743|
|      1.0|     2.0|[1.0,2.0]|272.12|285.24535397025454|
|      1.0|     2.0|[1.0,2.0]|274.42|285.24535397025454|
|      1.0|     2.0|[1.0,2.0]|267.16|285.24535397025454|
+---------+--------+---------+------+------------------+
only showing top 20 rows
```

```
from pyspark.ml.evaluation import RegressionEvaluator

rmseEvaluator = RegressionEvaluator(
    labelCol="price",
    predictionCol="prediction",
    metricName="rmse"
)

maeEvaluator = RegressionEvaluator(
    labelCol="price",
    predictionCol="prediction",
    metricName="mae"
)

r2Evaluator = RegressionEvaluator(
    labelCol="price",
    predictionCol="prediction",
    metricName="r2"
)

rmse = rmseEvaluator.evaluate(predTestDF)
mae = maeEvaluator.evaluate(predTestDF)
r2 = r2Evaluator.evaluate(predTestDF)

print("RMSE:", rmse)
print("MAE:", mae)
print("R2 Score:", r2)
```

```
RMSE: 22.31320924838175
MAE: 17.4912717458969
R2 Score: 0.9375834729983692
```

```
print("Intercept:", lrModel.intercept)
print("Coefficients:", lrModel.coefficients)
```

```
Intercept: 94.79645122400498
Coefficients: [31.597826083354285,79.42553833144765]
```

```
extra_features = [
    "bathrooms",
    "bedrooms",
    "beds",
    "accommodates",
    "review_scores_rating"
]

df_extra = df.na.drop(subset=extra_features + ["price"])

trainDF2, testDF2 = df_extra.randomSplit([0.8, 0.2], seed=42)

vecAssembler2 = VectorAssembler(
    inputCols=extra_features,
    outputCol="features"
)

vecTrainDF2 = vecAssembler2.transform(trainDF2)
vecTestDF2 = vecAssembler2.transform(testDF2)

lr2 = LinearRegression(
    featuresCol="features",
    labelCol="price",
    predictionCol="prediction"
)

lrModel2 = lr2.fit(vecTrainDF2)

predTestDF2 = lrModel2.transform(vecTestDF2)
```

```
predTestDF2.select(
    "bathrooms",
    "bedrooms",
    "beds",
    "accommodates",
    "review_scores_rating",
    "features",
    "price",
    "prediction"
).show()
```

```
+---------+--------+----+------------+-------------------+--------------------+------+------------------+
|bathrooms|bedrooms|beds|accommodates|review_scores_rating|            features| price|        prediction|
+---------+--------+----+------------+-------------------+--------------------+------+------------------+
|      3.0|     1.0| 1.0|           4|               96.1|[3.0,1.0,1.0,4.0,...|274.16|300.83402785585554|
|      2.5|     1.0| 1.0|           3|               78.0|[2.5,1.0,1.0,3.0,...|227.21|236.99455295707094|
|      1.0|     1.0| 1.0|           3|               72.0|[1.0,1.0,1.0,3.0,...|149.63|177.42904541647084|
|      4.0|     1.0| 2.0|           4|               85.0|[4.0,1.0,2.0,4.0,...|319.03|316.23405506984807|
|      1.0|     3.0| 4.0|           6|               70.2|[1.0,3.0,4.0,6.0,...|362.57|322.59991992498726|
|      4.0|     1.0| 1.0|           2|               77.5|[4.0,1.0,1.0,2.0,...|255.15| 270.3090427818572|
|      2.0|     1.0| 2.0|           4|               78.0|[2.0,1.0,2.0,4.0,...|217.86| 238.6605958493364|
|      4.0|     3.0| 4.0|           7|               73.4|[4.0,3.0,4.0,7.0,...|445.83|439.71952496758604|
|      3.0|     2.0| 2.0|           6|               73.8|[3.0,2.0,2.0,6.0,...|336.57| 339.2197260319317|
|      1.0|     3.0| 4.0|           7|               95.0|[1.0,3.0,4.0,7.0,...|362.99|382.65501789608265|
|      2.5|     2.0| 2.0|           5|               84.2|[2.5,2.0,2.0,5.0,...|302.99|328.02594989137225|
|      4.0|     3.0| 3.0|           7|               88.9|[4.0,3.0,3.0,7.0,...|452.07| 464.7687231141091|
|      4.0|     1.0| 2.0|           2|               74.0|[4.0,1.0,2.0,2.0,...|284.39| 267.4264546387973|
|      3.0|     4.0| 4.0|          10|               82.4|[3.0,4.0,4.0,10.0...|494.39| 514.2630933994375|
|      1.0|     1.0| 1.0|           2|               74.6|[1.0,1.0,1.0,2.0,...| 169.1|167.98770695328312|
|      4.0|     3.0| 4.0|           8|               70.9|[4.0,3.0,4.0,8.0,...|457.96| 449.3455851808026|
|      4.0|     2.0| 2.0|           5|               71.5|[4.0,2.0,2.0,5.0,...|377.29|353.04849017657557|
|      1.0|     2.0| 3.0|           6|               79.5|[1.0,2.0,3.0,6.0,...|272.12| 288.6886021730355|
|      1.0|     2.0| 3.0|           4|               90.2|[1.0,2.0,3.0,4.0,...|274.42| 279.9656214982474|
|      1.0|     2.0| 3.0|           4|               82.2|[1.0,2.0,3.0,4.0,...|267.16|265.18788149593854|
+---------+--------+----+------------+-------------------+--------------------+------+------------------+
only showing top 20 rows
```

```
rmse2 = rmseEvaluator.evaluate(predTestDF2)
mae2 = maeEvaluator.evaluate(predTestDF2)
r2_2 = r2Evaluator.evaluate(predTestDF2)

print("Original Model RMSE:", rmse)
print("Original Model MAE:", mae)
print("Original Model R2:", r2)

print("Improved Model RMSE:", rmse2)
print("Improved Model MAE:", mae2)
print("Improved Model R2:", r2_2)
```

```
Original Model RMSE: 22.31320924838175
Original Model MAE: 17.4912717458969
Original Model R2: 0.9375834729983692
Improved Model RMSE: 17.663806494054135
Improved Model MAE: 14.363809703970889
Improved Model R2: 0.9608849292314982
```